

Analysis of Published Papers

Individual Coursework

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Introduction

This report analyses a corpus of research papers published from 2000-2022 by four academic journals: the Journal of the Operational Research Society, the International Journal of Computer Mathematics, Applied Artificial Intelligence, and the Journal of Interdisciplinary Mathematics. There are 4385 papers in total, on a wide variety of topics, from varying fields and disciplining, providing opportunity to investigate overall themes and behaviours across these 4 related journals, but also the shifts in focus and structure that separate them.

There are four main foci of this report, posed as hypotheses and explorations. First, ‘Can papers be classified by abstract themes and structures?’, exploring whether abstracts have distinct topics or structural characteristics between journals. Second, ‘Have the themes and structures of abstracts shifted since 2015?’, exploring a potential shift of theme and structure across journals in 2015. Following into a section on the alignment of clusters with journals or identified topics. Finally, ‘Can citations be predicted for a newly published paper?’, using regression models to evaluate whether citations can be predicted using the paper’s abstract and metadata.

Data and Pre-processing

Data Description

Each of the 4385 published papers is an entry in the dataset. Each containing the metadata: title, the journal of publication, year published, page count, author, number of online views, citations, an altimetric score and then the abstract. The abstract is a concise summary of each papers content and findings, this was the primary source for most analysis undertaken as it provides insight into the paper, allowing thematic and lexical analysis.

Pre-Processing

The abstract texts were pre-processed to produce a clean corpus suitable for text-based analysis. The abstracts were then tokenized into individual words, which converted them into lowercase for a consistent representation. Common stop words were then filtered out using a standard stop word package, containing high-frequency everyday words. A small set of custom words were also filtered out after brief inspection of the top words, these were words like ‘data’, ‘method’ and ‘results’, which are very common in journals on maths, OR, and AI. Finally, tokens containing non-alphabetical characters were filtered out to reduce noise, such as page numbers referenced in abstract or years.

Can papers be classified by abstract themes and structures?

Lexical Characteristics by Journal

This section examines surface-level word usage across abstracts. Looking at the top words across all abstracts, then the frequency within each journal, providing an initial indication of whether the journal comparisons will have any success.

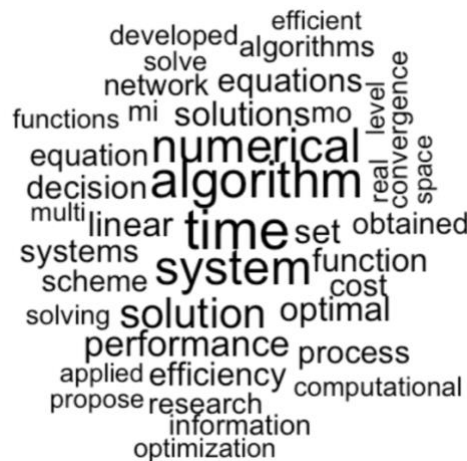


Figure 1: Word Cloud showing most common words across corpus

Figure 1 provides an overview of the most frequently occurring words across all abstracts. Initially the standout terms like numerical, algorithm, time and system were to be potentially added to the stop words. As it seemed their frequency was so high because they were basic words used across all journals equally, but as can be seen in figure 2, this was not the case.

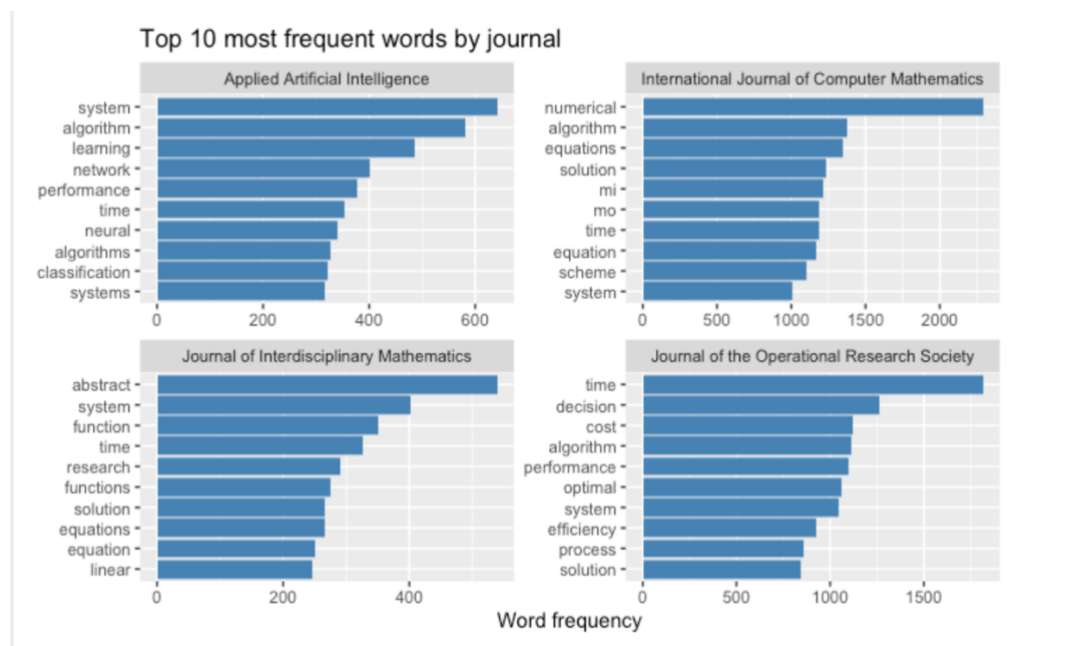


Figure 2: Top 10 most frequent words by journal

Examination of the most frequent words by journal reveals lexical patterns. For example, Applied Artificial Intelligence highlights words like ‘neural’, ‘network’ and ‘learning’, reflecting its focus on machine learning techniques and AI. Whilst, within Computer Mathematics, words like ‘numerical’, ‘equation’, and ‘solution’ show emphasis on numerical methods.

There is some overlap of high-frequency words, such as ‘time’ and ‘algorithm’ appearing across 3 journals. However, this doesn’t mean the journals are similar, instead the term can have different contextual meaning, such as optimising time in operation research, having time variables within a mathematic model, and computational runtime in artificial intelligence applications. This motivates further analysis past word frequency.

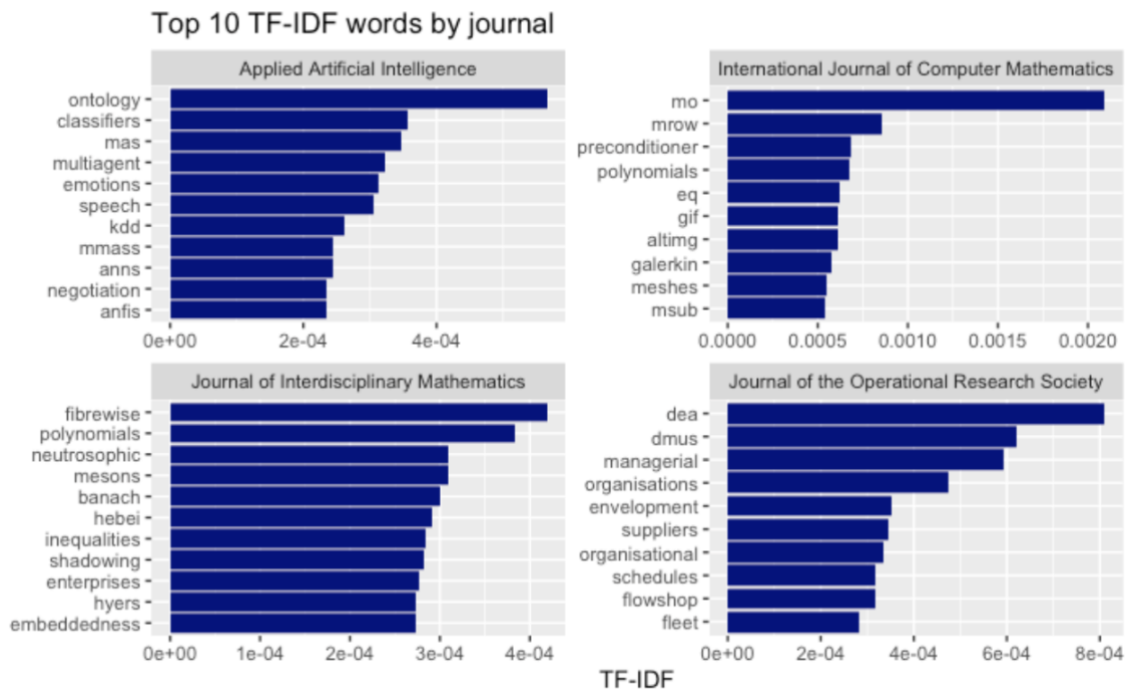


Figure 3: Top 10 TF-IDF words by journal

Figure 3 presents the highest TF-IDF weighted words for each journal, highlighting words that are frequent within a journal but rare within the others. There are now clearer differences between journals. Within Operational Research, terms like ‘suppliers’, ‘schedules’ and ‘fleet’ indicate a strong emphasis on logistics and supply-chain issues, consistent with usual focus of operational research.

In contrast, unusual terms like ‘mo’, ‘mrow’, ‘eq’ and ‘gif’ flagged for Computer Mathematics. These were initially thought to be discipline specific shorthand, but upon further inspection are

from latex-style formatting being captured within the abstracts. While these tokens don't carry much thematic meaning, their presence reflects some structural properties of the abstract, hence they were kept.

Thematic Structure of Abstracts

Whilst word frequency and TF-IDF can differences in word usage and distinctive terms, they don't capture the themes of the abstracts. To investigate if the abstracts can be grouped by topics, and assess how these topics vary across journals, topic modelling was applied using LDA.

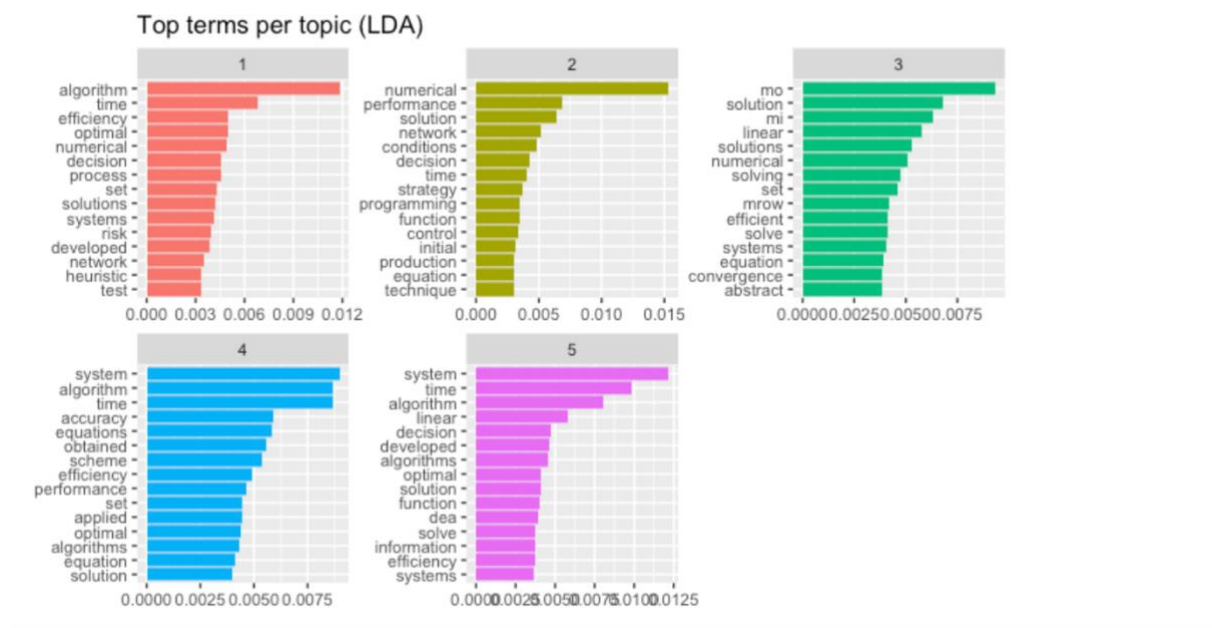


Figure 4: Top terms by topic

Several values for the number of topics were tested during model development, through inspection of the resulting distributions, five topics provided the most interpretable structure, although a four topic model still worked. Lower values of k tended to merge themes, and higher values of k split into topics that were very similar.

As shown in figure 4, topic 1 is characterised by top terms 'algorithm', 'time', 'efficiency' and 'decision', indicating a focus on algorithmic problems and computational methods. Topic 2 seems to focus on applied numerical and mathematical methods, while topic 3 related more to analytical and mathematical problem-solving. Topic 4 and 5 share similar terms like system and algorithm, however their emphasis varies. Topic 4 is more closely associated with system solutions and optimisation, whilst 5 is strongly linked to decisions and information.

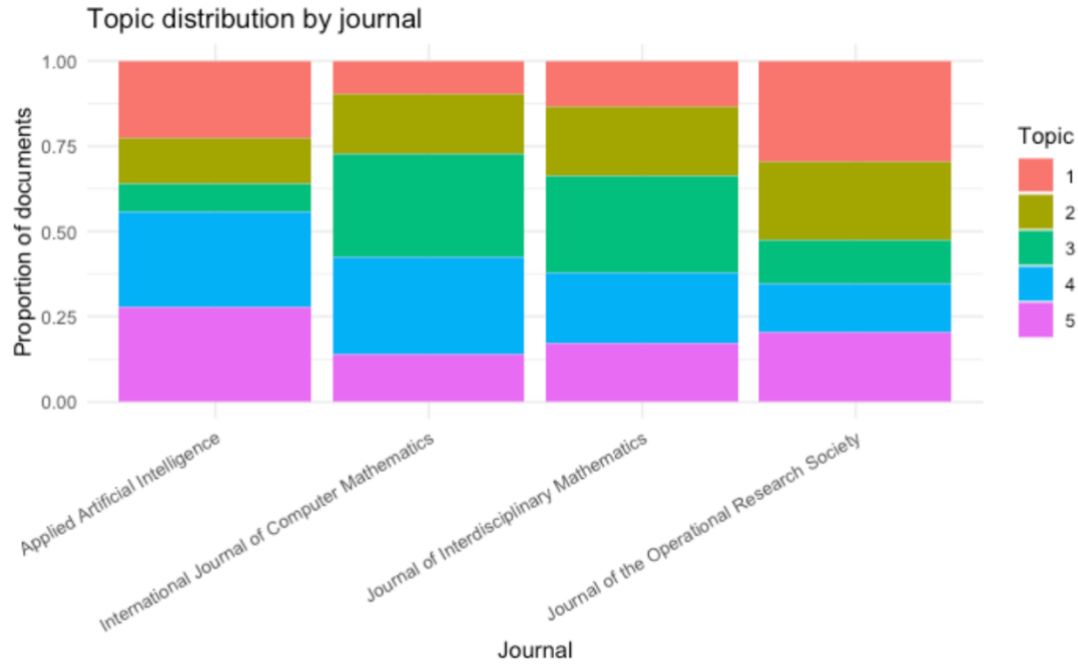


Figure 5: Distribution of LDA topics across journals

Although figure 5 shows that all journals contain all topics, the varying distributions are apparent. Applied Artificial Intelligence has a high proportion of topics 4 and 5 and, which aligns with the focus on systems and algorithms. Both AAI and OR journals have low proportions of topic 3, which focuses on numerical and mathematical methods, whilst the mathematics journals have a high proportion. In contrast, the mathematical journals have a low distribution of topic 1, the computational topic.

Overall, the topic proportions are reflective of what would be expected from each journal. This indicates that the topic modelling has captured meaningful differences in themes across the corpus.

Association Analysis

Association rule mining was applied separately across journals to find words that frequently occur together. The aim is to highlight local relationships between words to understand the concepts being discussed. For each journal, the top 15 rules ranked by lift were shown, with node size showing lift and colour showing confidence. Where lift showing how much more likely terms appear together than if they were independent, and confidence the likelihood of a term given the other.

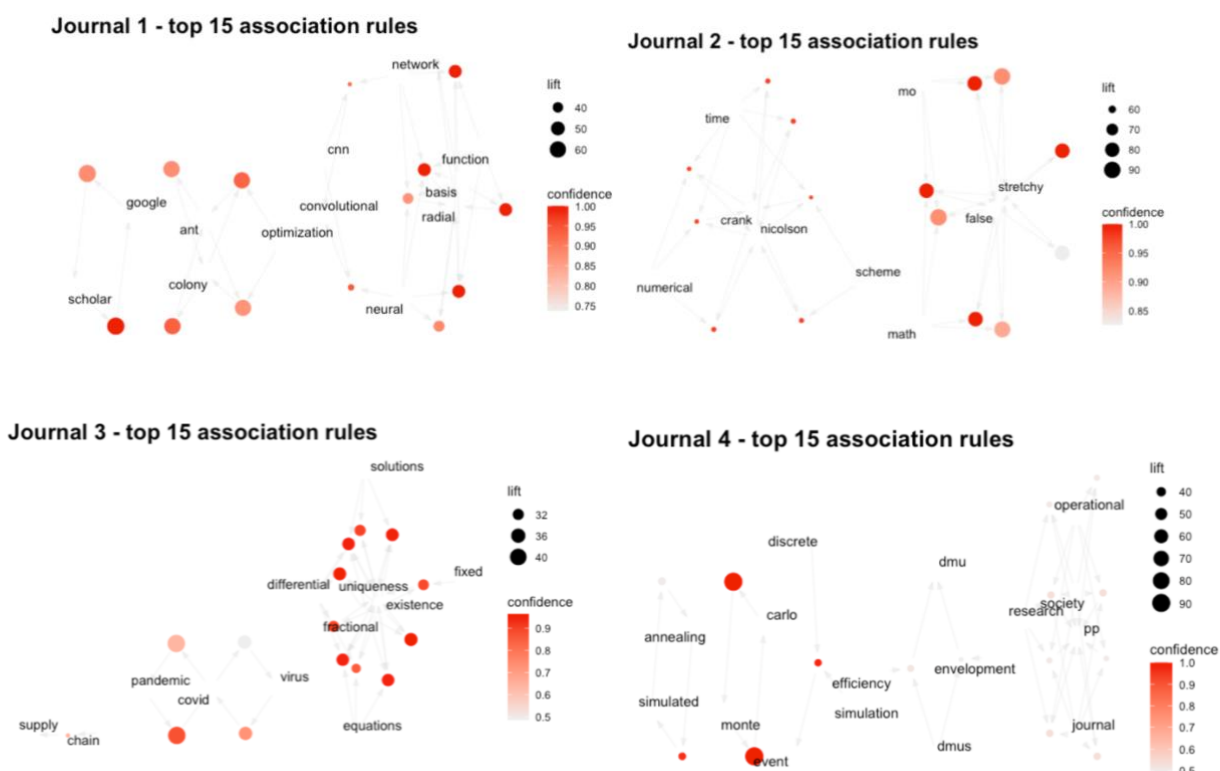


Figure 6: Top 15 association rules by journal

The association rules in figure 6 show clear differences in word co-occurrences across journals. In journal 1 (Applied Artificial Intelligence), there are strong associations between words relating to machine learning such as neural network, cnn, convolutional. Journal 2 (Computer Mathematics) had strong association between crank, numerical and Nicolson, which upon research relates to the ‘Crank-Nicolson Method’ a numerical technique. Journal 3 (Interdisciplinary Mathematics) shows very prominent associations between fractional, equations, differential, solutions, all of which are associated with mathematical research. Additionally, there is a set of terms relating to covid-19, demonstrating the applied research also. Finally, journal 4 (Operational Research) has multiple mention of simulation explicitly or through terms like monte and carlo. Overall, the association analysis highlights clear patters for each journal, that align with what would be expected of each journal.

Journal Classification

Having now identified distinct thematic and structural patterns across journals, this section aims to answer the question, ‘Can papers be classified by abstract themes and structures?’

Three text representation were tested, word frequency using the 1500 most frequent words in the corpus, a TF-IDF representation and a topic-based representation, using the document topic proportions from the LDA.

Both RandomForest and XGBoost models were evaluated for the word frequency representation, where XGBoost was shown to perform better. So, for the following experiments only XGBoost was used to save computational cost and time

Representation <chr>	Model <chr>	Accuracy <dbl>	Kappa <dbl>
BOW	RF	0.8081023	0.7157307
BOW	XGBoost	0.8432836	0.7726299
TF-IDF	XGBoost	0.8425729	0.7714885
Topics	XGBoost	0.4886283	0.2273526

Figure 7: Classification model performance

The comparison table shows that the XGBoost Model performed similarly for word-frequency (bag-of-words) and TF-IDF representations. Achieving accuracies above 0.84 and kappa values close to 0.8. Where the accuracy is proportion of correctly classified abstracts, whilst kappa accounts for correct occurrences by chance.

Have the themes and structures of abstracts shifted since 2015?

Exploration

From the outset, it was recognised that difficulties would arise when examining changes over time on a yearly basis, particularly in the context of classification. Treating each year independently can cause over-fragmentation, with papers published a few months apart sharing themes and structures being classified separately due to being in separate years. Similarly, grouping years into bins can cause issues at the boundaries, causing disruptive overlap.

For this reason, an exploratory analysis was done to investigate if there was clearer and more meaningful division. With hopes to notice a clear shift, simplify the task into a binary problem of pre and post shift.

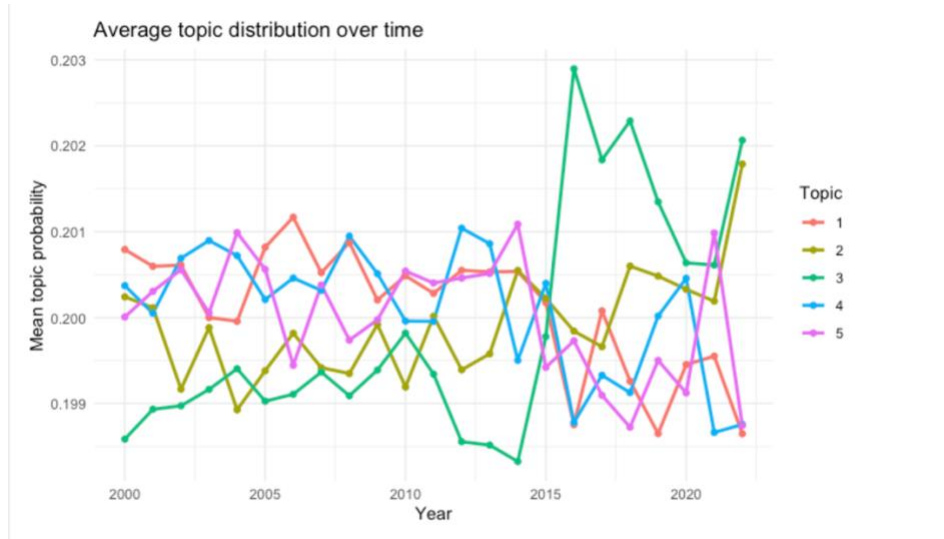


Figure 8: Average Topic distribution over time

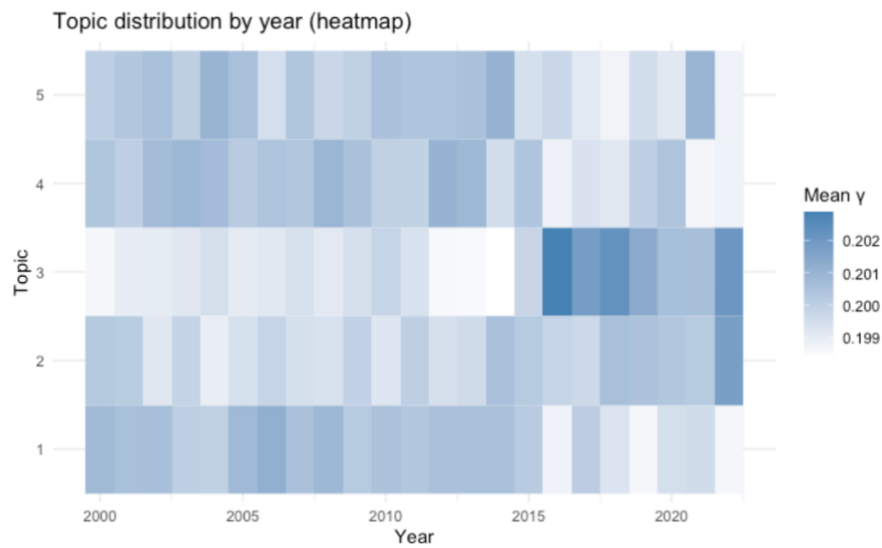


Figure 9: Heat map of topic distribution by year

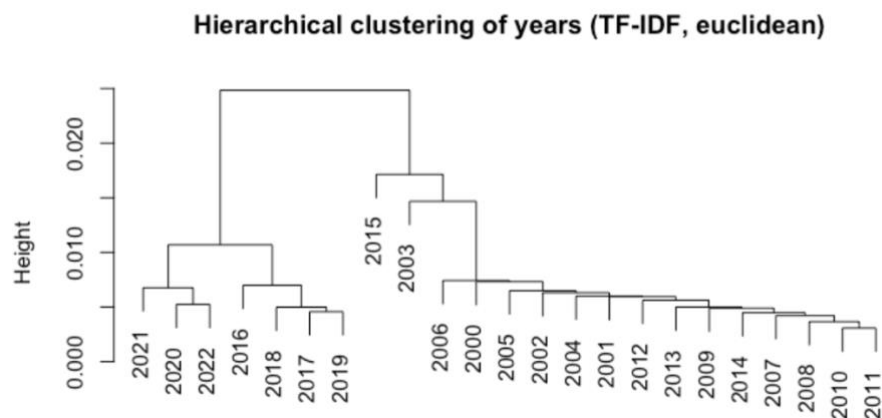


Figure 10: Hierarchical clustering of publication years based on TF-IDF representations

Figures 8-10 demonstrate lots of shifts in topics over time, and do not pose a definitive point where a shift takes place. However, a noticeable change in topics/themes takes place around 2015. In figure 8, topic 3 goes from having the lowest presence to the highest, crossing over the other in 2015. Additionally, in figure 9 topic 3 becomes significantly darker in 2015 onwards. Topic 3 relates to analytical and mathematical problem-solving, so this shift indicates a new focus on more complex and algorithmic approaches. Figure 10, also agrees with this conclusion, as it reveals a clear separation between 2015 and onwards. Additionally, 2015 is high on the dendrogram, meaning it is less similar to either cluster, indicating it is the transition year.

Association Analysis

Having split the abstracts into pre and post 2015 periods, association rule mining was used to investigate patterns of word co-occurrence within them both.

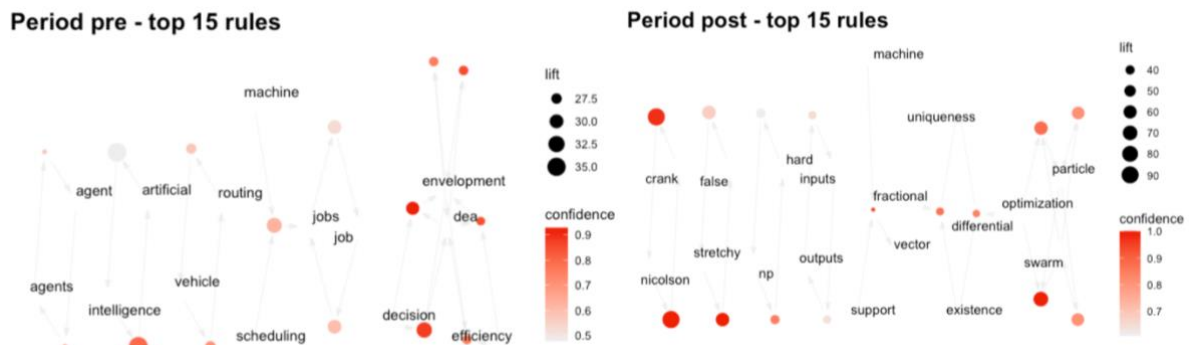


Figure 11: Top 15 association rules by period

Figure 11 clearly highlights the difference between the periods. In the pre-2015 period, strong associations are made regarding decision making and efficiency. There is mention of artificial intelligence, however it is in relation to practical applications like routing and scheduling. In the post-2015 period, there are associations between mathematical methods like crank-nicolson, as well as terms like differential, equation, fractional, vector, again showing the shift in focus to more mathematical techniques.

Can papers be classified by period using abstract themes and structures?

Following the exploration into thematic change over time, and confirmation of division year using association rules, a binary classification was performed.

Representation <chr>	Model <chr>	Accuracy <dbl>	Kappa <dbl>
TF-IDF	RF	0.7357016	0.3409569
TF-IDF	XGBoost	0.7211368	0.3880426

Figure 12: Classification model performance

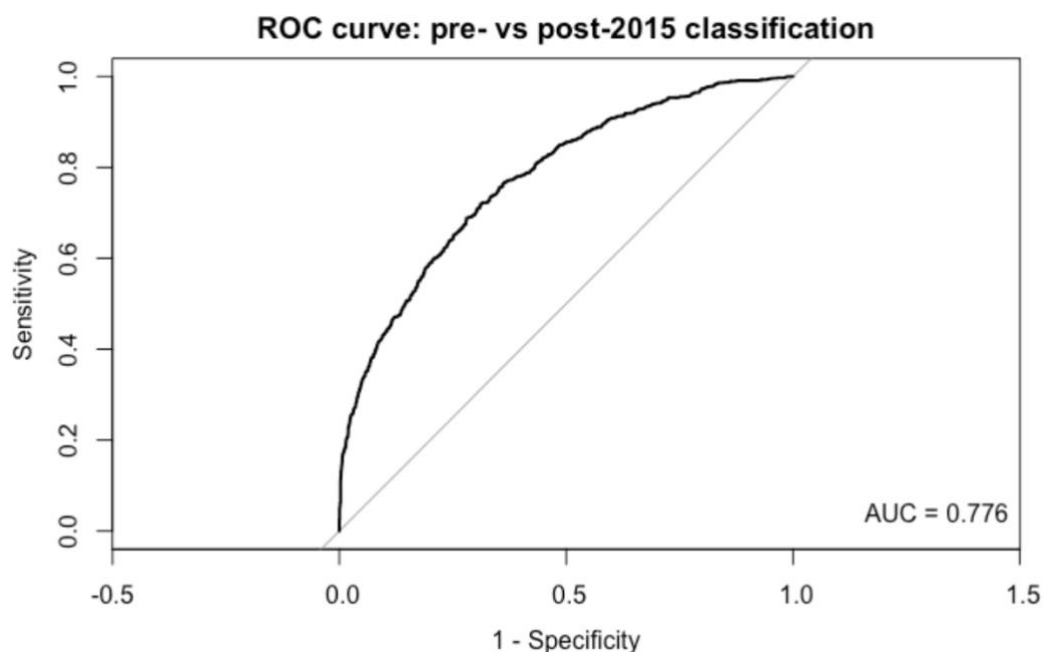


Figure 13: ROC/AUC curve for XGBoost

From the table, both models performed better than by chance with an accuracy around 0.72/73, and kappa values of 0.34 and 0.39. The ROC curve has an AUC value of 0.776. These values are quite weak, however they perform better than chance. This is expected when accounting for there not being a clean division, and the overlap across period.

Overall, the classification task as well as all prior explorative analysis points to a gradual structural and thematic shift around 2015.

Alignment of Clusters with Journals and Topics

Clustering was applied to the TF-IDF representations of abstracts to explore whether patterns in language align with themes or journal. Hierarchical clustering was used, and the number of clusters was chosen using a grid search between 2-8. The optimal value was $k=2$, selected using average silhouette width. Average silhouette width find how well observations are clustered, by comparing the distance to nearest observation within cluster to nearest observation in neighbouring cluster.

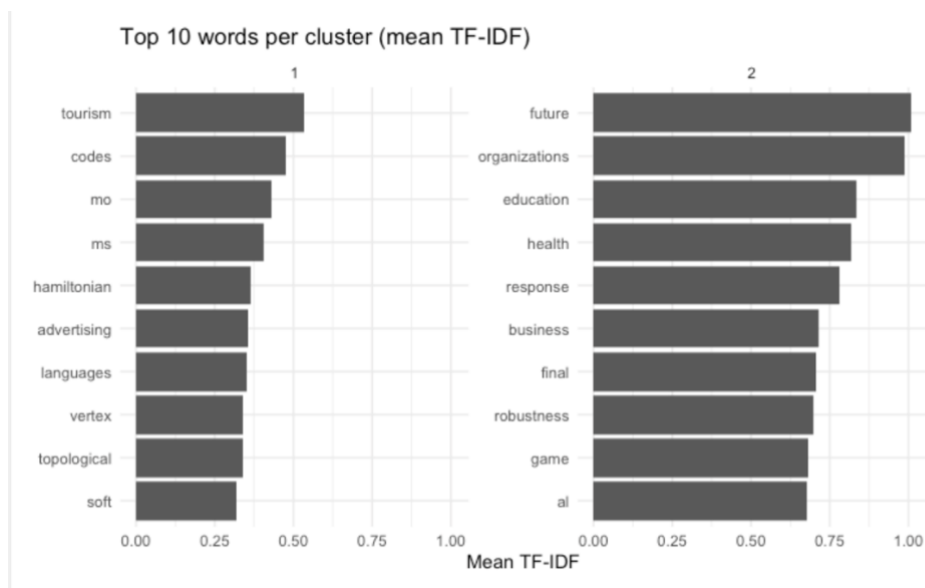


Figure 14: Top 10 words per cluster

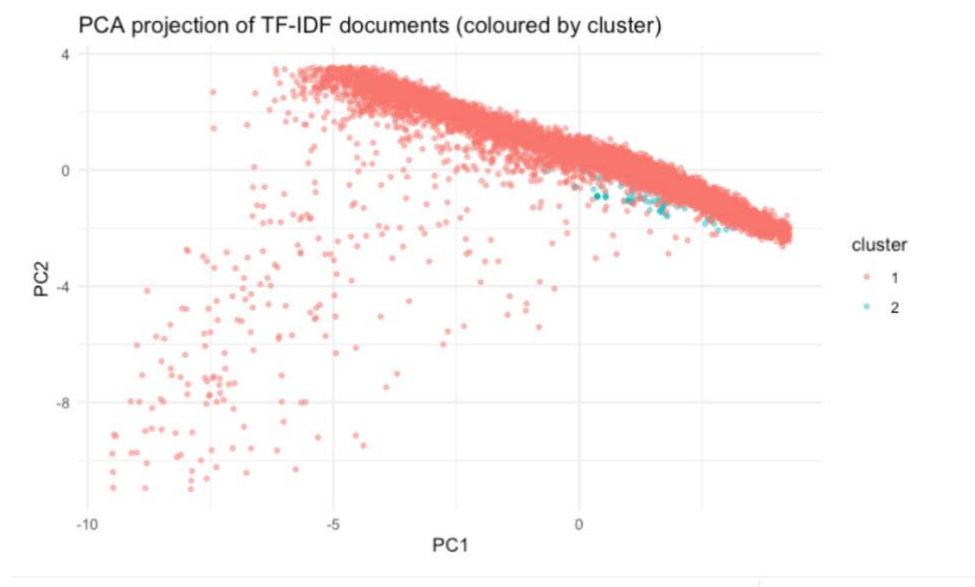


Figure 15: PCA projection of TF-IDF representations of abstracts, coloured by cluster

From figure 14, cluster 1 contains words like ‘tourism’, ‘codes’, ‘hamiltonian’, and ‘topological’, indicating an emphasis on mathematical formulation and modelling. Whilst cluster 2 contains words like ‘future’, ‘organisations’, ‘business’ and ‘educations’, indicating emphasis on real-world application with more impact.

From figure 15, cluster 1 contains most of the abstracts. This indicates that most papers share a broad language structure, whilst the small, contained cluster 2 represents a specialised group, capturing a more niche set of abstracts.

Can citations be predicted for a newly published paper?

This section investigates whether a regression model can be used to predict the number of citations on a new published paper, using journal data, themes and structure. A big issue with the training data is it is highly skewed, with lots of papers receiving no citations or very few, as can be seen in figure 16.

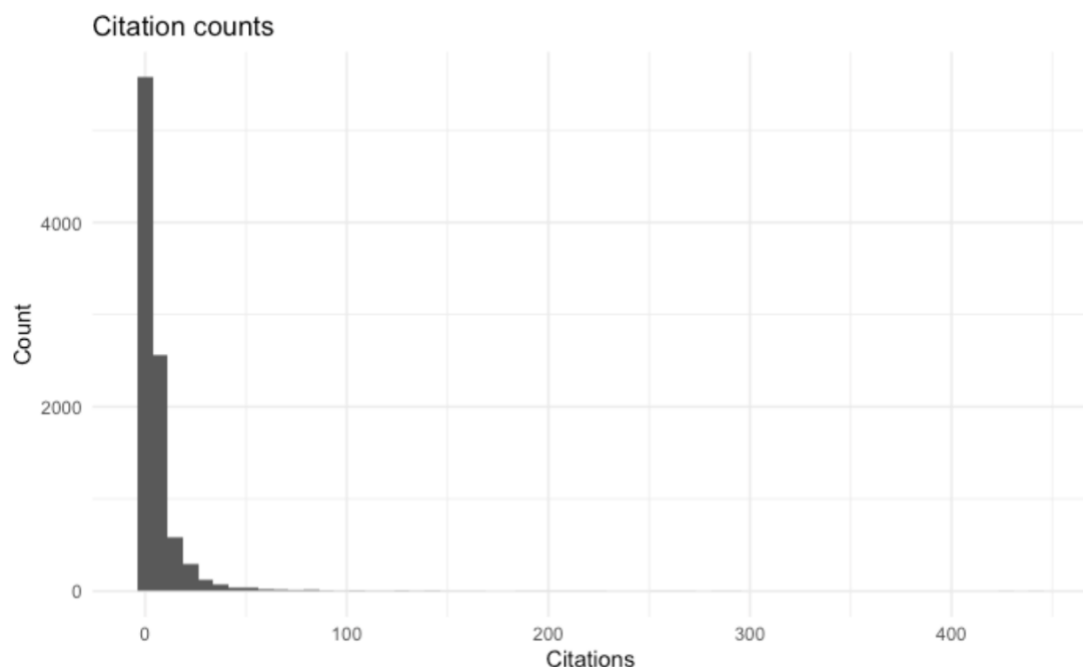


Figure 16: Skewed citation distribution

A multiple linear regression was first fitted, using abstract length, year, altimetric score, TF-IDF weight and topic proportions, to predict citation count. Performance on this first model was weak, with a RMSE of 16 and a R^2 of around 0.05. To address the skewness, a log transformed model of $\log(\text{citations} + 1)$ was used. Resulting in almost identical results.

An idea was posed to split the problem into two, first to predict if the paper would get any citations, followed by a separate model trained without the 0 values. The initial prediction on if a paper would get any citations had accuracy = 0.7669 and AUC = 0.6485, shown in figure 17. This is a weak model but still better than predicting by chance. However, when trying to fit a regression to the papers that are predicted to get at least 1 citation, the model performs poorly, with RMSE around 19 and $R^2 = 0.05$.

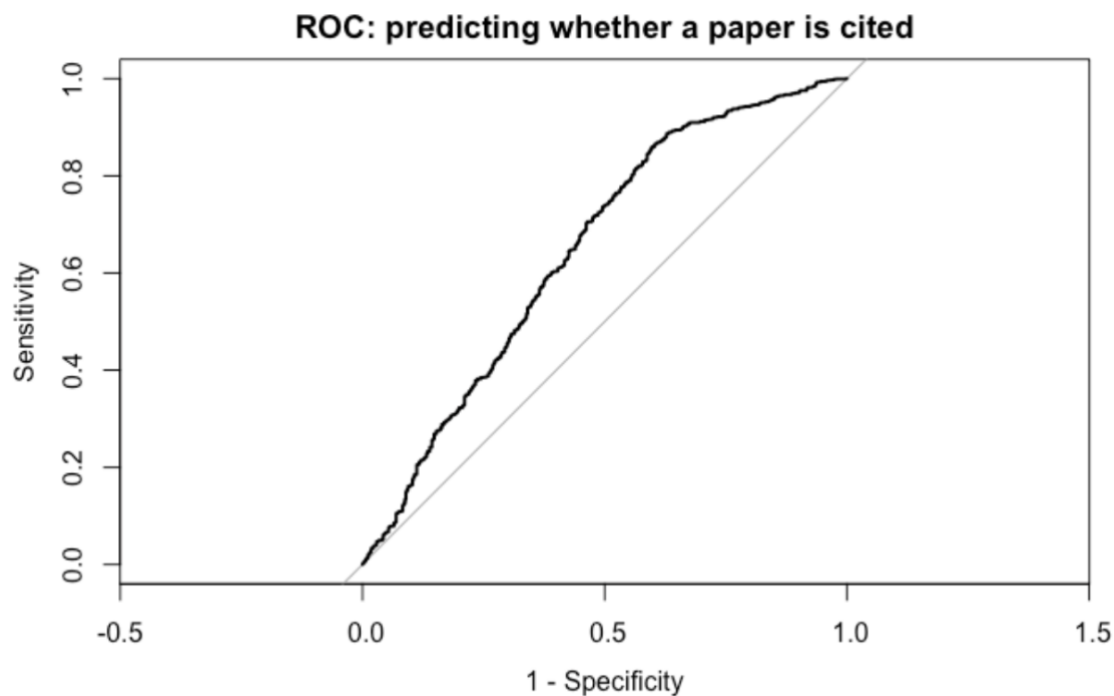


Figure 17: ROC curve for predicting if a paper will be cited

Overall, the regression results indicate that the papers metadata, the abstract theme and structure and may hold some signal towards potential citations, but they are insufficient to consistently and accurately predict citations. In reflection this may be due to the highly skewed data, or that citations can't fit to a model.